








Blockchain Applications

Lecture 6: Networks, Sentiment and Asset Classes



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Opening: How Are Crypto Assets Linked?

> *"Using a generalized vector autoregressive framework in which forecast-error variance decompositions are invariant to the variable ordering, we propose measures of both the total and directional volatility spillovers."*
>
> — *Diebold & Yilmaz (2012)*

- 19 coins, January 2021 – October 2026: the mean pairwise correlation of daily returns is 0.60
  - one common factor explains 68% of the variance on average (59% in October 2026)
- Spillovers among 8 coins: 73% of the forecast-error variance of a coin comes from the other coins
  - the rolling index rose from 65.2% (July 2021) to 83.5% (May 2026)
  - ETH and BTC are the largest net transmitters, DOGE the largest net receiver
- 13 Ethereum upgrades: large cumulative abnormal returns (CARs) of both signs; their mean is not distinguishable from zero
- BitMood, a sentiment index from Facebook posts: deep networks that use it forecast Bitcoin trading volume no better than a linear regression
  - test error (root mean squared error, RMSE): 0.1935 for the linear regression, 0.1930 for the best network

▸ Central chart: Spillovers through Time

Today's Question and Route

- **Question**: crypto assets do not move alone. How are they linked, are they an asset class of their own, and do protocol events and social media move prices?
- **Route**
  - networks: correlation trees, connectedness, Diebold–Yilmaz spillovers
  - are cryptos an asset class? factor analysis and classification
  - event studies: Ethereum upgrades
  - sentiment from social media: the BitMood index
- Data: End-of-Day Historical Data (EODHD) daily prices of 19 coins and 14 traditional assets, January 2021 – October 2026; Ether upgrade dates 2019–2025

Learning Outcomes

- Build a correlation distance, a minimum spanning tree and an absorption ratio, and interpret hubs and connectedness
- Read a Diebold–Yilmaz spillover table: total, TO, FROM and NET spillovers
- Explain how factor analysis and classification separate cryptos from other assets
- Run and interpret a market-model event study: abnormal return (AR), cumulative abnormal return (CAR), individual and cross-sectional tests
- Describe how a text-sentiment index is built and why trading back-tests need out-of-sample selection

Abbreviations (1/2)

ACF autocorrelation function
ADA Cardano
AI artificial intelligence
AIC Akaike information criterion
ALGO Algorand
AR absorption ratio (networks)
AR abnormal return (event studies)
ATOM Cosmos
BH Benjamini–Hochberg (adjusted p-values)
Bi-LSTM bidirectional LSTM
BMP Boehmer–Musumeci–Poulsen (test)
BNB BNB (Binance)
bp basis point (0.01%)
BTC Bitcoin
CAPM capital asset pricing model
CAR cumulative abnormal return
CNN convolutional neural network
CS compound score (sentiment)
CS cross-sectional (in t_{CS})
CSAD cross-sectional absolute deviation
CTE conditional tail expectation
DAI Digital Art Index
DOGE Dogecoin
DOT Polkadot
EIP Ethereum Improvement Proposal
EODHD End-of-Day Historical Data (data provider)
ETC Ethereum Classic
ETF exchange-traded fund
ETH Ether (Ethereum)
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FIGARCH fractionally integrated GARCH
FIL Filecoin
FinBERT BERT language model for financial text
FTX FTX crypto exchange (collapsed November 2022)

Abbreviations (2/2)

FX foreign exchange
GARCH generalised autoregressive conditional heteroskedasticity
GRU gated recurrent unit
HAC heteroskedasticity and autocorrelation consistent
KMO Kaiser–Meyer–Olkin (sampling adequacy)
LDA latent Dirichlet allocation
LINK Chainlink
LSTM long short-term memory (network)
LTC Litecoin
MA moving average
MST minimum spanning tree
NFT non-fungible token
n.s. not significant
PC1, PC2 first, second principal component
RF random forest
RV realised volatility
RMSE root mean squared error
S&P 500 Standard & Poor's 500 stock index
SC synthetic control
SCAR standardised cumulative abnormal return
sd standard deviation
SE standard error
SHAP Shapley additive explanations
SI reaction-intensity score of a post (BitMood)
SOL Solana
SR Sharpe ratio
SVM support vector machine
TRX Tron
USDT Tether (stablecoin)
VAR vector autoregression
VaR value at risk
vol volatility
WC engagement-weighted compound score (BitMood)
XMR Monero
XRP XRP Ledger token

Reading and Tools

- Asset classes: Pele, Wesselhöfft, Härdle, Kolossiatis & Yatracos (2023), *Are cryptos becoming alternative assets?*, European Journal of Finance
- Networks and spillovers: Mantegna (1999), Diebold & Yilmaz (2012); herding: Christie & Huang (1995), Chang, Cheng & Khorana (2000)
- Event studies: MacKinlay (1997), *Event studies in economics and finance*, Journal of Economic Literature 35(1); Lin, Găman, Wang & Pele (2026), *When does the market react to protocol innovation? Evidence from the Ethereum upgrade lifecycle*, working paper
- Sentiment: Conda, Găman, Bâg, Mazurencu-Marinescu-Pele, Pele & Härdle (2026), *BitMood: AI analysis of Bitcoin trends via Facebook emotions*, Digital Finance 8, article 42
- Video units on Quantinar: Dynamic Crypto Networks, Cross-exchange Crypto Risk, Reddit Sentiment and BTC Returns, Elon Musk's Twitter Feed, DAI Digital Art Index
- Software: Python libraries `statsmodels` (vector autoregression, VAR), `networkx` (networks, spanning trees), `scikit-learn` (principal components, clustering)

What Is a Link? One Word, Four Measures

measure	a link i–j means	direction, part of the distribution
tail link (Lecture 5)	coin j has a nonzero slope in the 5% Quantile Lasso of coin i	$j \to i$; lower tail only
correlation ρ_{ij}	returns move together on the same day	none (symmetric); whole distribution
MST edge	ρ_{ij} is among the $N-1$ strongest links that connect all coins	none; whole distribution
spillover $\tilde{\theta}_{ij}$ (Diebold–Yilmaz)	share of i's forecast-error variance due to shocks in j	$j \to i$; variance over H days

- In this lecture, "link" means a correlation or an MST edge; "spillover" means a Diebold–Yilmaz share
- None of the four measures shows that one coin *causes* another to move

From Correlations to a Network

- Nodes: 19 coins; edges: dependence of daily log returns, January 2021 – October 2026 (EODHD, 2 121 days)
- Pearson correlation ρ_{ij} becomes a metric distance (Mantegna, 1999):

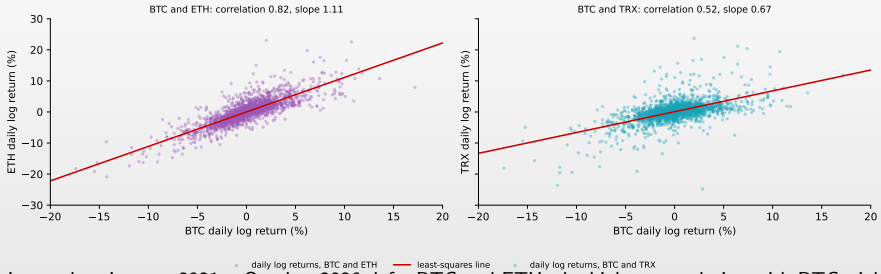
$$d_{ij} = \sqrt{2\,(1-\rho_{ij})}, \qquad \rho = 1 \Rightarrow d = 0, \quad \rho = 0 \Rightarrow d = \sqrt{2}$$

- Minimum spanning tree (MST): the $N-1$ edges that connect all coins with the smallest total distance

$$\mathrm{MST} = \arg \min_{T \text{ spanning tree}} \sum_{(i,j)\in T} d_{ij}$$

  - keeps the strongest links and removes the rest; a hub = a coin with many MST edges (high degree)
- Worked example: BTC–ETH, $\rho = 0.823$: $d = \sqrt{2 \cdot 0.177} = 0.595$; DOGE–TRX, $\rho = 0.349$: $d = 1.141$

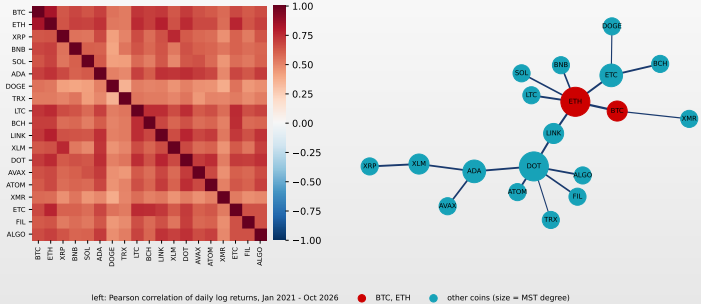
What a Correlation of 0.82 Looks Like



- Each dot is one day, January 2021 – October 2026; left: BTC and ETH, the highest correlation with BTC; right: BTC and TRX, the lowest
- BTC–ETH: $\rho = 0.82$; ETH moves by 1.11% on average per 1% move of BTC (least-squares slope); both coins move in the same direction on 82% of days
- BTC–TRX: $\rho = 0.52$, slope 0.67, same direction on 69% of days; $\rho^2 = 0.27$: BTC explains about a quarter of TRX's daily variance
- Distances: $d = 0.60$ for BTC–ETH and $d = 0.98$ for BTC–TRX; the MST prefers the short ones
- Source: own computation on EODHD daily prices of 19 coins, January 2021 – 22 October 2026

BA_L6_pairs

Correlation Matrix and Minimum Spanning Tree



left: Pearson correlation of daily log returns, Jan 2021 - Oct 2026 ● BTC, ETH ● other coins (size = MST degree)

- Mean pairwise correlation 0.60 (minimum 0.35, DOGE–TRX); correlation with BTC from 0.52 (TRX) to 0.82 (ETH)
- Two hubs: ETH (degree 6: BTC, BNB, SOL, LTC, LINK, ETC) and DOT (degree 6: ADA, ATOM, FIL, ALGO, TRX, LINK); BTC links only to ETH and XMR
- Source: own computation on EODHD daily prices of 19 coins, January 2021 – 22 October 2026

BA_L6_mst ▶ Interactive

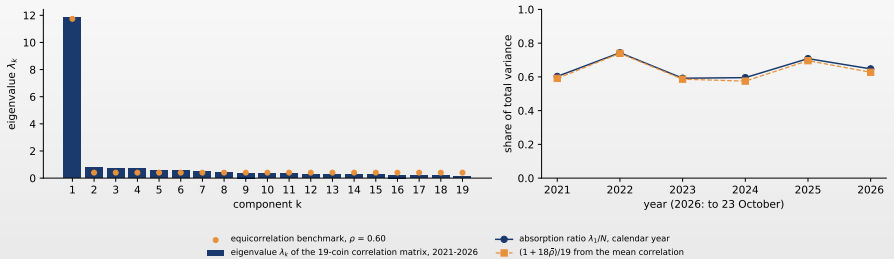
The Absorption Ratio: a Worked Example

- Principal components of a correlation matrix of N coins: uncorrelated portfolios of the standardised returns
  - eigenvalue λ_k = variance of the k-th component; $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$ and $\sum_k \lambda_k = N$
- Absorption ratio (AR): share of the total variance taken by the first component, between $1/N$ and 1

$$\mathrm{AR} = \frac{\lambda_1}{\sum_{k=1}^{N} \lambda_k} = \frac{\lambda_1}{N}$$

- Worked example: $N = 19$ coins, every pairwise correlation equal to ρ
  - $\lambda_1 = 1 + (N-1)\rho$; the other 18 eigenvalues equal $1 - \rho$
  - $\rho = 0.60$: $\lambda_1 = 1 + 18 \cdot 0.60 = 11.8$, $\mathrm{AR} = 11.8/19 = 0.62$
  - $\rho = 0.85$: $\lambda_1 = 16.3$, $\mathrm{AR} = 0.86$
  - $\rho = 0$: $\mathrm{AR} = 1/19 = 0.05$, no common factor
- A high AR means one common factor drives most coins: diversification across coins is weak

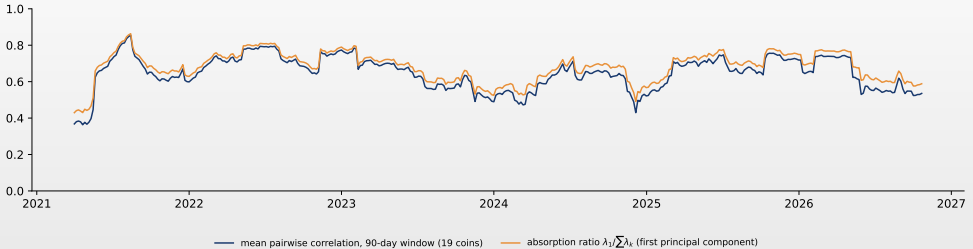
One Factor Takes Almost Everything



- Left: the 19 eigenvalues of the full-sample correlation matrix. $\lambda_1 = 11.84$, the next one 0.79; only λ_1 exceeds 1
- The equicorrelation formula with the mean correlation $\bar{\rho} = 0.60$ predicts $\lambda_1 = 11.74$ and 0.40 for each of the others
- Right: the absorption ratio by calendar year, from 0.59 (2023) to 0.74 (2022, the year of Terra and FTX)
- $(1 + 18\bar{\rho})/19$ is within 0.021 of the absorption ratio in every year: the mean correlation already tells how strong the common factor is
- Source: own computation on EODHD daily prices of 19 coins, January 2021 – 22 October 2026

BA_L6_eigen ▶ Interactive

How Connected, and When?



- 90-day windows of daily log returns, 19 coins; the last window ends on 22 October 2026
- Mean pairwise correlation: from 0.36 (April 2021) to 0.85 (mid-August 2021, after the May crash); 0.54 in the last window
- Absorption ratio AR_t: 0.68 on average over all windows, 0.59 in the last window
- When one factor explains 80% of the variance, "diversifying across coins" buys almost nothing
- Source: own computation on EODHD daily prices of 19 coins, January 2021 – 22 October 2026

BA_L6_connect ▶ Interactive

Spillovers: Diebold–Yilmaz

- VAR(p) on N return series, $\mathrm{r}_t = \sum_{l=1}^{p} \Phi_l \mathrm{r}_{t-l} + \varepsilon_t$, moving-average (MA) form $\mathrm{r}_t = \sum_{h \geq 0} A_h \varepsilon_{t-h}$
- Generalised forecast-error variance decomposition (Pesaran–Shin), horizon H:

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathrm{e}_i' A_h \Sigma \mathrm{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathrm{e}_i' A_h \Sigma A_h' \mathrm{e}_i}, \qquad \tilde{\theta}_{ij} = \frac{\theta_{ij}}{\sum_j \theta_{ij}}$$

  - $\tilde{\theta}_{ij}$: share of coin i's forecast-error variance due to shocks in coin j; independent of variable ordering
- Indices, in % ($\tilde{\theta}_{ij}$ is a share between 0 and 1, so each sum is multiplied by 100):

$$S = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}, \quad \mathrm{TO}_j = 100 \sum_{i \neq j} \tilde{\theta}_{ij}, \quad \mathrm{FROM}_i = 100 \sum_{j \neq i} \tilde{\theta}_{ij}, \quad \mathrm{NET}_i = \mathrm{TO}_i - \mathrm{FROM}_i$$

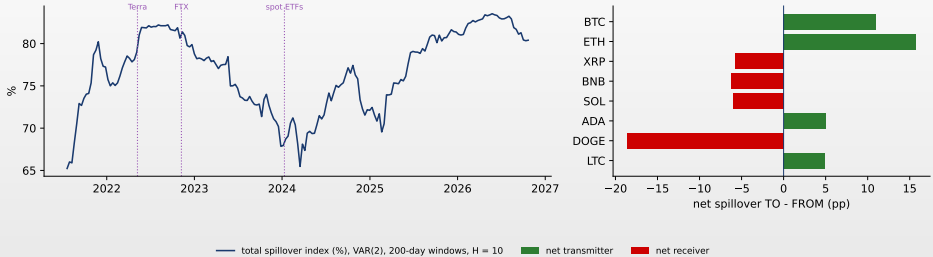
  - TO_j: what coin j sends to all others; FROM_i: what coin i receives from all others

Spillover Table, 8 Coins, 2021–2026

receiver ↓ / source →	BTC	ETH	XRP	BNB	SOL	ADA	DOGE	LTC	FROM
BTC	23.3	16.0	9.4	10.6	10.0	11.3	7.3	12.0	76.7
ETH	15.3	22.5	9.5	10.5	10.9	11.8	6.9	12.5	77.5
XRP	11.2	11.8	28.1	9.0	8.6	13.0	5.9	12.4	71.9
BNB	12.7	13.3	8.9	28.4	10.4	10.0	4.9	11.3	71.6
SOL	12.1	13.8	8.5	10.5	28.3	11.6	5.7	9.4	71.7
ADA	12.1	13.2	11.5	8.7	10.3	25.0	7.2	12.0	75.0
DOGE	11.4	11.1	7.3	6.0	7.1	10.4	36.4	10.2	63.6
LTC	12.8	14.0	11.0	9.9	8.3	11.9	7.1	25.0	75.0
TO	87.7	93.2	66.1	65.4	65.7	80.0	45.0	80.0	$S = 72.9$
NET	+11.0	+15.7	−5.8	−6.3	−6.0	+5.0	−18.6	+5.0	

- VAR(5) by Akaike information criterion (AIC), $H = 10$ days. Row = receiver of shocks; column = source
- Reading a cell: row BTC, column ETH = 16.0: 16% of BTC's 10-day forecast-error variance is due to ETH shocks (ETH → BTC)
- Worked check: $\mathrm{NET}_{ETH} = 93.2 - 77.5 = +15.7$; S = mean of the FROM column = 72.9%: a highly connected system
- ETH and BTC are the largest net transmitters. DOGE is the largest net receiver (−18.6), although it receives the least (FROM 63.6): it sends far less (TO 45.0)
- Source: own computation on EODHD daily prices of 8 coins, January 2021 – October 2026

Spillovers through Time



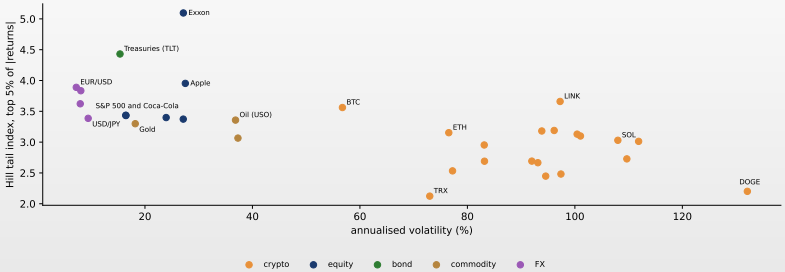
- Rolling VAR(2), 200-day windows: from 65.2% (July 2021) to 83.5% (May 2026); 80.4% in the last window, ending 22 October 2026
- Spillovers rose into Terra and FTX, fell during 2023, and climbed after the spot exchange-traded funds (ETFs)
- The rise after the ETFs is consistent with more common (institutional) flows; the chart alone does not identify the cause
- Source: own computation on EODHD daily prices of 19 coins, January 2021 – 22 October 2026

BA_L6_spill ▶ Interactive

Classifying Assets by Their Statistical "Phenotype"

- Idea of Pele et al. (2023): describe each asset by the stylised facts of its returns, then ask whether cryptos form their own group
- Six features per asset, from daily log returns 2021–2026: annualised volatility, skewness, log kurtosis, Hill tail index (top 5%), lag-1 autocorrelation (ACF) of $|r_t|$, 1% quantile in sd units
- 33 assets: 19 coins, 6 equities, Treasuries, gold, silver, oil, 4 foreign exchange (FX) rates
- Each asset is measured on its own trading days (coins every day, traditional assets Monday–Friday): a stale holiday close would add false zero returns
- Each feature is standardised across the 33 assets (mean 0, sd 1), so that no feature dominates through its units
- Principal components: first (PC1) explains 62.7%, second (PC2) 18.5%
  - PC1 loadings (weights of the standardised features): vol 0.40, skew 0.39, kurtosis 0.46, tail index -0.44, ACF$|r|$ 0.46, 1% quantile 0.27: a "wildness" factor
- k-means, $k = 3$:
  - cluster 1: all 14 traditional assets *and* BTC, ETH, LTC, LINK
  - cluster 2: 14 other coins; cluster 3: DOGE alone (skewness 5.9)
- Source: own computation on EODHD daily prices of 19 coins and 14 traditional assets, January 2021 – October 2026

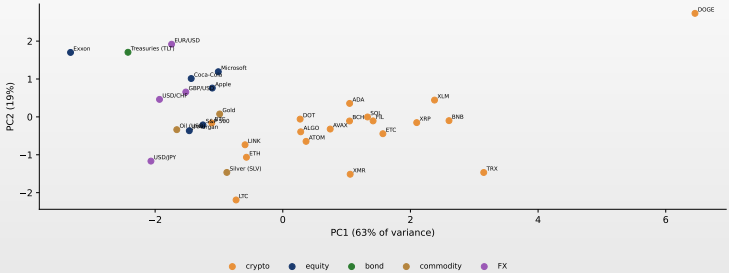
Volatility and Tails of 33 Assets



- Two of the six features, daily log returns 2021–2026: annualised volatility (horizontal) and Hill tail index of the top 5% of $|r_t|$ (vertical; lower = heavier tail)
- Volatility separates the groups: every coin is above 56% a year (BTC 57%, DOGE 132%; median coin 95%); every traditional asset is below 38% (silver)
- Tails overlap: coins 2.12 (TRX) to 3.66 (LINK), traditional assets 3.07 (silver) to 5.10 (Exxon); BTC (3.56) has a lighter tail than the S&P 500 (3.44)
- Source: own computation on EODHD daily prices of 19 coins and 14 traditional assets, January 2021 – October 2026

BA_L6_features

The Map of Assets



- Bitcoin (vol 57%, Hill tail index 3.56) sits next to the S&P 500 (vol 16%, tail index 3.44): volatility is one feature of six, and on the other five BTC resembles equities
- Shape only (volatility dropped, same method): BTC, ETH, LTC, LINK, DOT, AVAX, ATOM and ALGO join the cluster of all 14 traditional assets; 10 altcoins form a second cluster, DOGE a third
- Most altcoins remain a separate cloud; DOGE is an outlier of its own
- Source: own computation on EODHD daily prices of 19 coins and 14 traditional assets, January 2021 – October 2026

BA_L6_classes ▶ Interactive

Worked Example: Where Bitcoin Sits on PC1

- Step 1, standardise each feature across the 33 assets: $z = (x - \text{mean})/\text{sd}$
  - volatility: $(56.7 - 62.3)/38.9 = -0.14$; Hill tail index: $(3.56 - 3.22)/0.61 = +0.57$
  - the other four z-values of BTC: skewness -0.26, log kurtosis -0.69, ACF$|r|$ -0.50, 1% quantile -0.61
- Step 2, weight each z by its PC1 loading and add: $\mathrm{PC1} = \sum_k w_k z_k$

	vol	skew	kurtosis	tail index	ACF$\lvert r\rvert$	$Q_{1\%}$	sum
loading w_k	0.40	0.39	0.46	-0.44	0.46	0.27	
z_k of BTC	-0.14	-0.26	-0.69	0.57	-0.50	-0.61	
$w_k z_k$	-0.06	-0.10	-0.31	-0.25	-0.23	-0.16	-1.11

- Step 3, compare: S&P 500 -1.25, ETH -0.57, DOGE $+6.46$; BTC ranks 23rd of 33 on the "wildness" factor
- Kurtosis and the tail index contribute most: on shape, BTC is calmer than the average asset of the sample, which is why it sits next to equities
- Source: own computation on EODHD daily prices of 19 coins and 14 traditional assets, January 2021 – October 2026

Reading the Indicators of Pele et al. (2023)

- Stable $\alpha \in (0, 2]$: tail parameter of a stable distribution fitted to *all* returns; $\alpha = 2$ is the Normal, $\alpha < 2$ means infinite variance; smaller = heavier tails
- Hill tail index (Lecture 2): estimated from the largest returns only; usually 2–4 for financial returns. Stable α and Hill index come from different models and parts of the distribution: their values are not comparable
- Hurst exponent $H \in (0, 1)$: memory in returns; 0.5 = none, above 0.5 = persistent trends, below 0.5 = mean reversion
- FIGARCH $d \in [0, 1]$: memory in volatility; $d = 0$ is GARCH (shocks fade geometrically), larger d = shocks fade more slowly (hyperbolically)
- Factor loading: correlation between an indicator and a factor, from -1 to 1; an indicator "belongs" to the factor it loads on most
- Kaiser–Meyer–Olkin (KMO) $\in [0, 1]$: share of the indicators' correlation that is common to all of them; above 0.8 = data well suited to factor analysis
- Nagelkerke $\tilde{R}^2 \in [0, 1]$: pseudo-R^2 of a logit, rescaled so that 1 = perfect classification

Pele et al. (2023): Cryptos as a Separate Asset Class

- Pele, Wesselhöfft, Härdle, Kolossiatis & Yatracos (2023): 906 assets (234 cryptos, 635 stocks, 13 FX, 17 commodities, 5 bonds, 2 real-estate indices), daily, January 2014 – November 2020
- 24 indicators per asset, on expanding windows: variance, skewness, kurtosis, stable α and γ, left and right quantiles and conditional tail expectations (CTE) at 0.5–5%, $\rho(1)$, Hurst exponent, fractionally integrated GARCH (FIGARCH) d
- Factor analysis (principal components, Varimax; Kaiser–Meyer–Olkin (KMO) $= 0.92$): 3 factors explain 88%
  - **tail** factor 76% (quantiles, CTEs, variance, stable parameters); **memory** 6% (Hurst, $\rho(1)$, d); **moment** 6% (skewness, kurtosis)
- Logit $P(\text{crypto}) = \Lambda(\beta_0 + \beta_1 F)$: tail $\hat{\beta}_1 = 15.45$ (standard error (SE) 4.44), Nagelkerke $\tilde{R}^2 = 0.992$; moment -0.74 ($\tilde{R}^2 = 0.12$); memory n.s.
- Support vector machine (SVM) on two factors: 99.56% accuracy, only BTC and USDT misclassified; k-means (6 clusters): 98.45%
- Conclusion: a separate asset class, "mainly due to the tail factor"; BTC lies at the border

What Separates Cryptos? Averages by Class (Pele et al., 2023)

	Cryptos	Stocks	Bonds	FX	Commodities	Real estate
$\sigma^2 \cdot 10^3$	26.442	0.400	0.745	0.028	0.388	0.146
stable α	1.342	1.602	1.460	1.722	1.662	1.696
$Q_{1\%}$	−0.366	−0.054	−0.057	−0.013	−0.050	−0.034
$\mathrm{CTE}_{1\%}$	−0.564	−0.085	−0.097	−0.017	−0.074	−0.052
skewness	0.686	−0.560	−0.829	−0.937	−0.495	−1.142
FIGARCH d	0.553	0.289	0.545	0.407	0.426	0.510

- Daily returns in decimals, 2014–2020: crypto $Q_{1\%} = -0.366$ is a 1% daily quantile of -36.6%, the average over 234 cryptos
- Crypto variance is about 60 times that of stocks; stable $\alpha = 1.34$ against 1.60: heavier tails within the stable model
- Separation by projection pursuit: without stable α no direction separates cryptos perfectly; with it, 181 618 directions do
- Average 1% value at risk (VaR) of an equal-weight portfolio (historical): classical 3.01%, cryptos 16.34%, mixed 4.37%
- 2021–2026 (previous slides): on standardised features, and on shape features alone, BTC and ETH group with traditional assets, consistent with "BTC at the border"; most altcoins stay apart

The Market-Model Event Study

- Normal return from a market model estimated on $[-250, -31]$ days before the event:

$$R_t = \alpha + \beta R_{m,t} + u_t, \qquad \mathrm{AR}_t = R_t - \hat{\alpha} - \hat{\beta} R_{m,t}, \qquad \mathrm{CAR}(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} \mathrm{AR}_t$$

- One event: $t = \mathrm{CAR}(\tau_1, \tau_2) / \sqrt{L\,\hat{\sigma}_u^2}$, $L = \tau_2 - \tau_1 + 1$; simplified: it ignores the estimation error of $\hat{\alpha}, \hat{\beta}$, which is small for a long window (220 days: $|t|$ overstated by 0.7–1.7% for the Ether upgrades)
- Many events $e = 1, \ldots, N$: cross-sectional (CS) test, and a simplified Boehmer–Musumeci–Poulsen (BMP) test on standardised CARs (SCAR, without the forecast-error correction)

$$t_{CS} = \frac{\overline{\mathrm{CAR}}}{s_{\mathrm{CAR}}/\sqrt{N}}, \qquad \mathrm{SCAR}_e = \frac{\mathrm{CAR}_e}{\hat{\sigma}_{u,e}\sqrt{L}}, \qquad t_{BMP} = \frac{\overline{\mathrm{SCAR}}}{s_{\mathrm{SCAR}}/\sqrt{N}}$$

- Sign test: number of positive CARs $\sim \mathrm{Bin}(N, 0.5)$ under H_0
- Application: Ether against an equal-weighted index of 11 other coins, daily; 13 upgrade families (Istanbul 2019 to Fusaka 2025), two dates each: formal schedule and implementation
- Event clustering: the CS, BMP and sign tests assume independent events; for Ether, two schedule dates are 2 days apart (Istanbul, Muir Glacier), and 17 of 26 estimation windows contain another event day

Worked Example: the Merge, Step by Step

- Implementation date: 15 September 2022. Market model fitted on days $[-250, -31]$: $\hat{\alpha} = 0.064\%$, $\hat{\beta} = 1.054$, residual sd $\hat{\sigma}_u = 2.08\%$
- Step 1, abnormal return each day: $\mathrm{AR}_t = R_t - \hat{\alpha} - \hat{\beta} R_{m,t}$ (returns in %)

day	ETH R_t	index $R_{m,t}$	AR_t	calculation
-1 (14 Sep)	3.36	1.66	1.54	$3.36 - 0.064 - 1.054 \times 1.66$
0 (15 Sep)	-10.51	-2.85	-7.56	$-10.51 - 0.064 + 1.054 \times 2.85$
$+1$ (16 Sep)	-2.70	2.22	-5.10	$-2.70 - 0.064 - 1.054 \times 2.22$

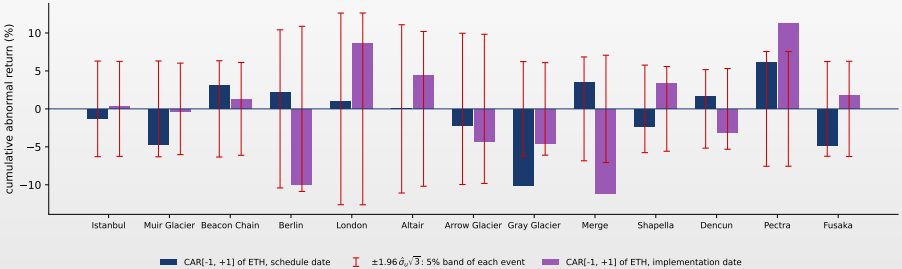
- Step 2, cumulate: $\mathrm{CAR}[-1, 1] = 1.54 - 7.56 - 5.10 = -11.13\%$
- Step 3, standard error over $L = 3$ days: $\hat{\sigma}_u \sqrt{3} = 2.08 \times 1.732 = 3.61\%$
- Step 4, test: $t = -11.13/3.61 = -3.09$; $|t| > 1.96$, so this single CAR is significant at 5%
- On the day itself the market model predicts $0.064 - 1.054 \times 2.85 = -2.94\%$ for ETH; the rest of the -10.51%, that is -7.56%, is abnormal
- Source: own computation on EODHD daily prices of Ether and 11 other coins; upgrade dates 2019–2025

CARs around 13 Upgrades

implementation	CAR$[-1,1]$	t
Istanbul (Dec 2019)	0.4%	0.12
Muir Glacier	−0.3%	−0.10
Beacon Chain	1.3%	0.41
Berlin	−9.9%	−1.79
London (EIP-1559)	8.7%	1.35
Altair	4.4%	0.84
Arrow Glacier	−4.3%	−0.85
Gray Glacier	−4.6%	−1.48
Merge (Sep 2022)	−11.1%	−3.09
Shapella	3.3%	1.17
Dencun	−3.1%	−1.16
Pectra (May 2025)	11.3%	2.92
Fusaka (Dec 2025)	1.7%	0.54

- Cross-section, implementation:
  - $\overline{\mathrm{CAR}}[-1,1] = -0.18\%$, $t_{CS} = -0.10$ (p 0.92), $t_{BMP} = -0.20$
  - 7 of 13 positive (sign test p = 1.00)
  - $\overline{\mathrm{CAR}}[0,5] = -0.21\%$ (p 0.94)
- Schedule dates: $\overline{\mathrm{CAR}}[-1,1] = -0.59\%$ (p 0.63)
- Implementation minus schedule, paired: $+0.41\%$ (p 0.84)
- Merge and Pectra have large CARs of opposite sign; no common revaluation of Ether is detectable in these windows
- Source: own computation on EODHD daily prices of Ether and 11 other coins; upgrade dates 2019–2025

Which CARs Are Significant?



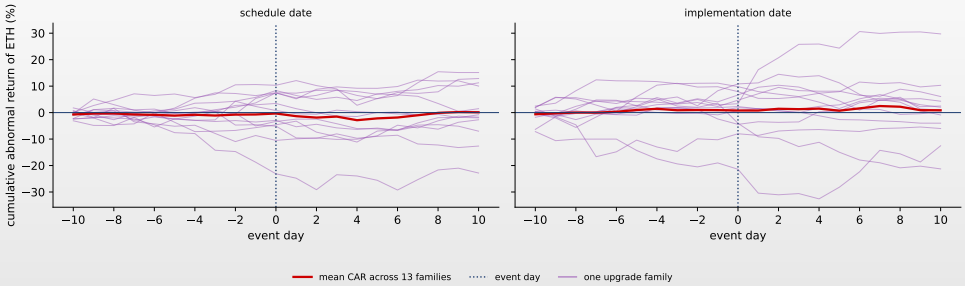
- Bars: CAR$[-1,+1]$ of each upgrade, schedule (blue) and implementation (purple) dates. Red whiskers: $\pm 1.96\,\hat{\sigma}_u\sqrt{3}$, the 5% band of that event
- The band differs by event because $\hat{\sigma}_u$ does: $\pm 5.2\%$ (Dencun schedule) to $\pm 12.6\%$ (London, in the volatile 2021 market)
- Outside the band: Merge and Pectra (implementation), Gray Glacier (schedule): 3 of 26 dates, against 1.3 expected by chance at 5%
- Berlin (-9.9%) is almost as large as the Merge but stays inside its wider band
- Source: own computation on EODHD daily prices of Ether and 11 other coins; upgrade dates 2019–2025

BA_L6_cars ▶ Interactive

Worked Example: Testing 13 CARs Together

- The 13 implementation CARs$[-1, 1]$ in %:
  $0.39, -0.31, 1.28, -9.91, 8.67, 4.35, -4.27, -4.59, -11.13, 3.32, -3.14, 11.25, 1.73$
- Step 1, mean: $\overline{\mathrm{CAR}} = -2.36/13 = -0.18\%$
- Step 2, spread across events: $s_{\mathrm{CAR}} = 6.50\%$; standard error of the mean $= 6.50/\sqrt{13} = 1.80\%$
- Step 3, cross-sectional test: $t_{CS} = -0.18/1.80 = -0.10$; critical value $t_{\mathbf{12,\,0.975}} = 2.18$; p = 0.92: we do not reject a zero mean
- Step 4, sign test: 7 of 13 CARs are positive; under H_0 the count is $\mathrm{Bin}(13, 0.5)$ with mean 6.5, so 7 is as central as possible: p = 1.00
- Two individual CARs are significant (Merge -11.1%, Pectra $+11.3\%$), but they cancel: large, opposite-signed reactions are compatible with a zero average effect
- The test needs s_{CAR} from the cross-section, not $\hat{\sigma}_u$: it allows each upgrade to have its own true effect
- Source: own computation on EODHD daily prices of Ether and 11 other coins; upgrade dates 2019–2025

Abnormal Returns around the Event Day



- Individual paths fan out in both directions; the mean path stays near zero
- Estimated betas of Ether to the coin index: 0.75 (Dencun) to 1.20 (Fusaka); residual sd 1.5–3.7% a day
- Source: own computation on EODHD daily prices of Ether and 11 other coins; upgrade dates 2019–2025

BA_L6_event ▶ Interactive

Lin et al. (2026): Attention, Activity, Volatility, Not Price

- Lin, Gǎman, Wang & Pele (2026): 13 upgrade families; outcomes beyond returns; cross-asset difference-in-differences and synthetic control (SC) against donor coins

$$\hat{\delta}_{e,s} = \Delta y_{e,s,ETH} - \frac{1}{J_e}\sum_{j=1}^{J_e} \Delta y_{e,s,j}, \qquad \hat{y}^{SC}_{e,t} = \sum_j w_{e,j}\, y_{j,t},\ w_{e,j} \geq 0,\ \textstyle\sum_j w_{e,j} = 1$$

- Exact sign-flip tests at family level, Benjamini–Hochberg adjusted p-values (BH p)

outcome (log points)	estimate	BH p	meaning
Google attention, schedule day	0.184	0.034	search starts at scheduling
Google attention, implementation day	0.350	0.004	+42%
active addresses, implementation (synthetic control)	−0.145	0.020	−13.5%
median hourly log realised volatility (RV), implementation hours	0.316	0.007	+37% volatility
jump share, bipower variation (24 h)	≈ 0	0.99	no lasting change
paired CAR, implementation minus schedule	−0.0098	raw p 0.375	no repricing detected

- The event goes with more attention and volatility and fewer active addresses
- A non-significant CAR does not prove that upgrades are "priced in", and fewer addresses do not show that "users pause": both are hypotheses consistent with these estimates

BitMood: Building a Sentiment Index

- Conda, Găman, Bâg, Mazurencu-Marinescu-Pele, Pele & Härdle (2026): 120 000 English Facebook posts on Bitcoin (5% monthly sample of about 2.3 million, CrowdTangle), 3 January 2015 – 15 October 2023
- FinBERT gives each post probabilities S^+, S^0, S^-; compound score (CS) $\mathrm{CS}_i = S_i^+ - S_i^-$
- Engagement-weighted compound score (WC), daily, with T_i = total interactions of post i:

$$P_d^{\pm,0} = \frac{\sum_{i \in d} S_i^{\pm,0} T_i}{\sum_{i \in d} T_i}, \qquad \mathrm{WC}_d = \frac{P_d^+ - P_d^-}{P_d^+ + P_d^- + P_d^0} \in [-1, 1]$$

- Reaction-intensity score (SI) of a post (weights from the literature): $\mathrm{SI}_i = \sum_k (\log N_{i,k} + 1)\, w_k$, $N_{i,k}$ = number of reactions of type k; like 0.1, comment 0.05, share 0.2, love/wow/haha 0.5, sad/angry -0.5
  - $\log 0$ is undefined and the article does not say how posts with no reactions of a type are treated; $\log(1 + N_{i,k})$ is a common fix
- Topics (latent Dirichlet allocation, LDA): investment 42.8%, trading and blockchain 24.6%, gaming 14.3%, social-media links 13%, business 5.3% of tokens

BitMood: a Contrarian Trading Rule

- Contrarian rule (after Preis et al.): $\Delta f(t, \delta t) = f(t) - \frac{1}{\delta t} \sum_{k=1}^{\delta t} f(t-k)$; buy when sentiment fell, sell when it rose
- Sharpe ratio (SR): mean return of the strategy divided by the standard deviation of its returns; Lecture 7 annualises it with $\sqrt{365}$
- Best configuration, test period 1 January – 15 October 2023: $\delta t = 3$ days, compound sentiment, cost 10^{-5}
  - cumulative return 1.321 and Sharpe ratio 0.210
  - the article does not say whether the return is simple, log or a wealth multiple, or whether the SR is daily or annualised
  - a cost of 10^{-5} as a fraction of the amount traded is 0.1 basis point (bp) per trade, far below typical exchange fees
- Caveat: the best of many window–feature–cost combinations (7 windows $\times$ 6 costs $\times$ several sentiment series) was chosen on the same 2023 period on which it is scored

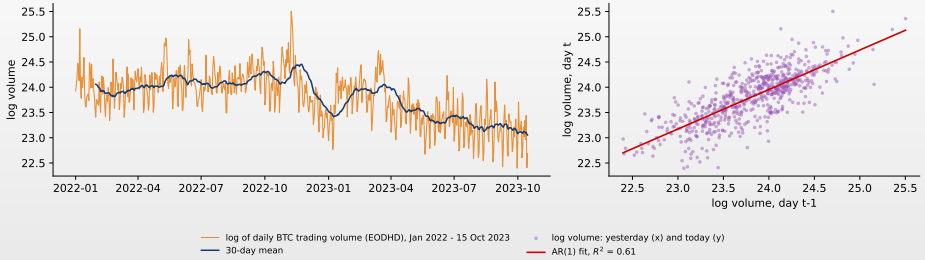
BitMood: Volume Forecasts

- Daily Bitcoin trading volume, January 2022 – October 2023; root mean squared error (RMSE) of scaled volume for linear regression, random forest (RF), long short-term memory (LSTM), convolutional (CNN) and gated recurrent unit (GRU) networks:

	Lin. reg.	RF	stacked LSTM	CNN	LSTM-CNN	GRU	Bi-LSTM
validation	0.1423	0.1767	0.2323	0.2179	0.2501	0.1840	0.1701
test	0.1935	0.2355	0.2453	0.2578	0.2766	0.1959	0.1930

- Select on validation, report on test: the linear regression has the lowest validation RMSE (0.1423, bold) and a test RMSE of 0.1935, 0.0005 above the best test value (Bi-LSTM)
- Lagged volume and negative sentiment dominate the Shapley additive explanations (SHAP) of the linear model
- Every model in the table uses sentiment: the table does not show that sentiment improves on lagged volume alone, and SHAP does not replace that test
- No evidence here on returns

The Benchmark BitMood Must Beat: Lagged Volume



- Log of daily Bitcoin trading volume (EODHD), 1 January 2022 – 15 October 2023, the BitMood volume sample; 653 days
- Autocorrelation of log volume: 0.78 at lag 1, 0.74 at lag 7 (weekly pattern: log volume is 0.33 below its mean on Saturdays and 0.28 on Sundays), 0.39 at lag 30
- An autoregression of order 1, AR(1), on yesterday's log volume explains 61% of the variance of today's (R^2); root mean squared error (RMSE) in logs: 0.50 with the sample mean, 0.33 with "same as yesterday", 0.31 with the AR(1)
- Sentiment must lower the error *beyond* this benchmark; the BitMood table does not report that comparison
- Source: own computation on EODHD daily Bitcoin volume, 1 January 2022 – 15 October 2023

BA_L6_volume

Before Trusting a Back-Test: the Minimum Standard

- Three periods, in time order:
  - training: estimate the parameters
  - validation: choose the model, window, threshold or cost setting (Lecture 4: alarm threshold chosen on validation)
  - test: score the chosen configuration once; nothing is changed after seeing it
- A benchmark without the new information, scored on the same test days
  - for BitMood: the same volume model without the sentiment features
  - report the RMSE difference with a confidence interval (for example, a Diebold–Mariano test, Lecture 7)
- Report the number of configurations tried
- Lecture 7 applies this standard to machine-learning forecasts of Bitcoin and adds the corrections for trying many rules

Summary

- Crypto returns form a dense network: mean correlation 0.60; ETH and DOT are the hubs of the minimum spanning tree; one factor explains 68% of the variance on average (59% in October 2026)
- Spillovers: 73% of forecast-error variance comes from other coins; ETH and BTC are the largest net transmitters, DOGE the largest net receiver; the index reached 83.5% in May 2026
- Cryptos are a separate asset class through their tails (Pele et al., 2023: tail factor 76%, SVM 99.6%); on standardised features, and on shape alone, BTC and ETH group with classical assets in 2021–2026
- Ethereum upgrades: large CARs of both signs, mean CAR not distinguishable from zero; Lin et al. (2026) report effects on attention, activity and volatility
- Sentiment models forecast volume about as well as a linear regression; whether sentiment adds to lagged volume is not tested; trading claims need selection on validation data

Interpretation Exercises

1. The MST shows BTC with only two links while ETH has six. Does this mean BTC is less important for the market? Use the spillover table in your answer.
2. The absorption ratio rises from 0.45 to 0.85. What happens to the benefit of holding ten coins instead of one?
3. In the spillover table, $\mathrm{NET}_{DOGE} = -18.6$. Explain what this number measures and what it does not prove.
4. The Merge has CAR$[-1, 1] = -11.1\%$ ($t = -3.09$) and Pectra $+11.3\%$ ($t = 2.92$), while the mean over 13 upgrades is -0.18% (p 0.92). What can and cannot be concluded?
5. A sentiment strategy reports a Sharpe ratio of 0.21 in 2023. Which two questions would you ask before using it?

⊡ Next: Lecture 7, machine learning, artificial intelligence (AI) and their limits; exam review

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- Do Investors Follow the Crowd? The CSAD Test
- Herding Is Not a Constant
- Why Not Just the Median Price?
- Worked Example: Composition Bias in a Boom (simulated)

Do Investors Follow the Crowd? The CSAD Test

- Cross-sectional absolute deviation of returns around the market return $R_{m,t}$:

$$\mathrm{CSAD}_t = \frac{1}{N}\sum_{i=1}^{N} |R_{i,t} - R_{m,t}|$$

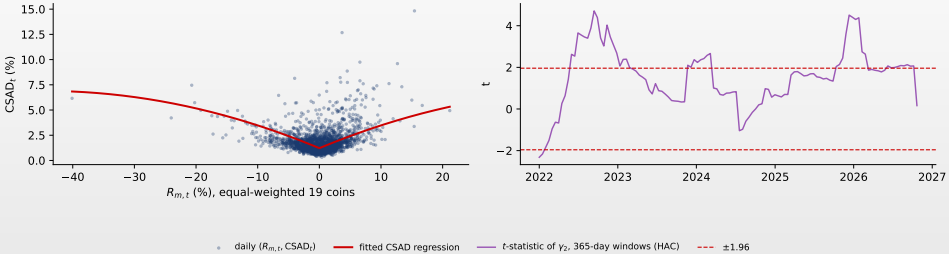
- Rational pricing (capital asset pricing model, CAPM): dispersion rises linearly with $|R_{m,t}|$. Herding: in big moves, coins move together and dispersion rises *less*

$$\mathrm{CSAD}_t = \alpha + \gamma_1 |R_{m,t}| + \gamma_2 R_{m,t}^2 + \varepsilon_t, \qquad H_0 : \gamma_2 = 0, \quad \text{herding: } \gamma_2 < 0$$

- Estimates for 19 coins, January 2021 – October 2026, equal-weighted R_m, $n = 2\,121$, heteroskedasticity and autocorrelation consistent (HAC(10)) errors:

sample	$\hat{\gamma}_1$ (t)	$\hat{\gamma}_2$ (t)	R^2
all days	0.254 (10.76)	−0.0028 (−2.88)	0.248
up days ($R_m > 0$)		0.0003 (0.03)	
down days		−0.0001 (−0.12)	

Herding Is Not a Constant



- Full sample: $\hat{\gamma}_2 < 0$ significant; the fitted curve is concave (vertex at $|R_m| = \hat{\gamma}_1/2|\hat{\gamma}_2| \approx 45\%$), and the curvature rests on the 16 days with $|R_m| > 15\%$
- 365-day windows: γ_2 significantly negative in only 1.7% of windows, significantly *positive* (anti-herding) in 39%
- Interpretation: herding in crypto is an episode, not a trait; a full-sample coefficient can hide the opposite behaviour most of the time
- Source: own computation on EODHD daily prices of 19 coins, January 2021 – October 2026

BA_L6_csad

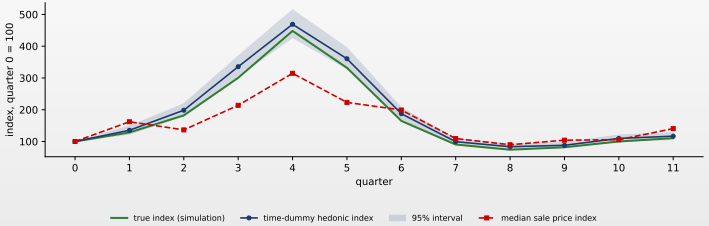
Why Not Just the Median Price?

- Each non-fungible token (NFT) or digital artwork is unique and trades rarely; the works sold this quarter differ from those sold last quarter
- Median sale price mixes two things: price changes and *composition* changes
- Time-dummy hedonic regression on all sales n in quarters $t = 0, \ldots, T$:

$$\ln P_n = \sum_{t=1}^{T} \delta_t D_{n,t} + \mathrm{x}_n'\boldsymbol{\beta} + \varepsilon_n, \qquad I_t = 100\, e^{\hat{\delta}_t}$$

  - x_n: characteristics (rarity, artist, size, collection); $\hat{\delta}_t$: quality-adjusted log price level
- Used for the Digital Art Index (Quantinar unit) and for real estate and art markets in general

Worked Example: Composition Bias in a Boom (simulated)



- 3 600 simulated sales, 12 quarters; in the boom (quarters 2–5) more low-rarity, unknown-artist works are sold
- Hedonic regression recovers the characteristics (rarity 2.52, known artist 1.20; true 2.5 and 1.2) and the index: RMSE 0.09 in logs
- Median index: RMSE 0.25; at the peak it shows 315 instead of 448 (30% too low): the boom looks smaller because cheaper works flood the market

BA_L6_hedonic